LEVEL 10 · PART V

Observability

10

Goal: Master the three pillars - metrics, logs and traces - and know what to monitor.

RECOMMENDED TIME	TOPICS
3-4 weeks	11

PRACTICE / CAPSTONE

Add Prometheus, Grafana, Loki, OpenTelemetry and Sentry to the sample app and build a golden-signals dashboard with an alert.

Topics

#	Topic	What you will learn
1	The Three Pillars: Metrics, Logs and Traces **p.1921**	monitoring vs observability, when to use which signal
2	Prometheus and PromQL **p.1933**	pull model, scrape config, exporters, rate(), histogram_quantile, recording/alert rules
3	Grafana Dashboards **p.1949**	data sources, panels, variables, dashboards as code, provisioning
4	Counters, Gauges, Histograms and Summaries **p.1961**	when to use each, instrumenting an app, cardinality
5	Structured Logging **p.1975**	JSON logs, levels, context fields, what not to log
6	Log Aggregation with Loki **p.1989**	Promtail/Alloy, labels, LogQL, retention
7	Correlation IDs and Request IDs **p.2005**	propagation across services, linking logs to traces
8	Distributed Tracing **p.2021**	spans, traces, context propagation, sampling, Jaeger/Tempo
9	OpenTelemetry **p.2035**	SDK, collector, auto-instrumentation, OTLP, vendor neutrality
10	Sentry **p.2055**	error tracking, releases, source maps, alert rules, vs metrics
11	What to Monitor: Golden Signals, RED and USE **p.2073**	CPU memory disk network, request rate, error rate, latency, DB connections, queues

#	Topic	What you will learn
★	Level capstone project p.2091	Add Prometheus, Grafana, Loki, OpenTelemetry and Sentry to the sample app and build a golden-signals dashboard with an alert.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

